



# Lecture 7: PN junction under forward bias; Diodes

ECE 340 – Electronic Circuits I Sec. 1

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(textbook sections 3.5-4.2)



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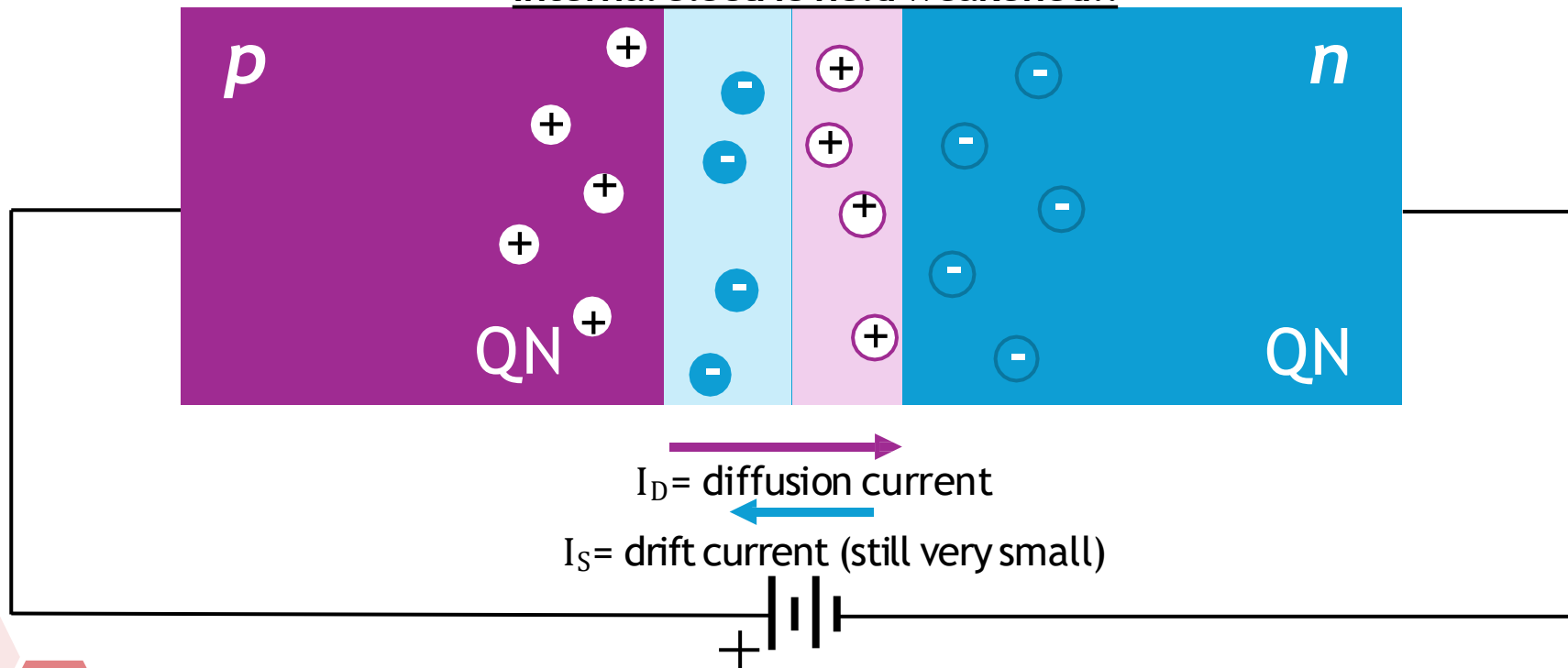
# Previously...

## Forward bias

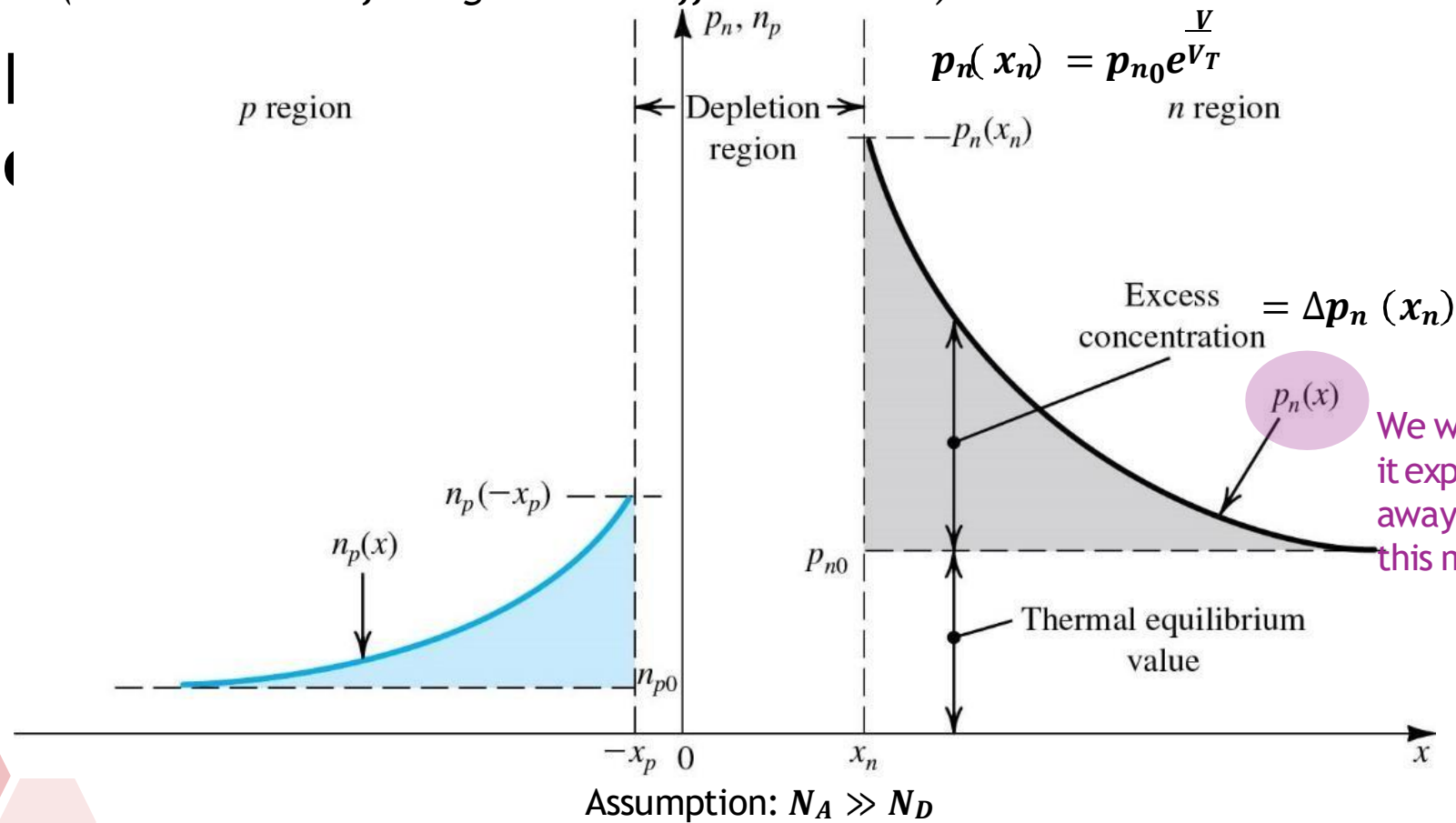
We increase # of holes in the p region

We increase # of electrons in the n region

Internal electric field weakened!!



(aka distribution of charges due to diffusion current)



We want to calculate this - it exponentially decays away from  $x_n$ , so we know this much

# FWD Bias - Hole diffusion current in the n-region

Holes in the n-region:

$$p_n(x) = p_{n_0} + \Delta p_n(x_n) e^{-\frac{x-x_n}{L_p}}$$

where  $L_p$  is the diffusion length for holes in [cm]

$x_n$  is the edge of the depletion region - get closer to this and the # of holes increases

(the distance a hole travels on the n-side before recombining with an electron)

$$p_n(x) = p_{n_0} + p_{n_0} \left( e^{\frac{V}{V_T}} - 1 \right) e^{-\frac{x-x_n}{L_p}} \quad (\text{for } x \geq x_n)$$

$$\Delta p_n(x_n) = p_n(x_n) - p_{n_0} = p_{n_0} \left( e^{\frac{V}{V_T}} - 1 \right)$$

Holes in the n-region move by diffusion, creating a diffusion current density of:

$$J_{D,p} = -qD_p \frac{dp_n(x)}{dx} \left[ \frac{A}{\text{cm}^2} \right]$$

# Total current through the junction

$$J(x) = J_n(-x_p) + J_p(x_n)$$

$I = A[J_n(-x_p) + J_p(x_n)]$ , where  $A$  is the junction cross-section

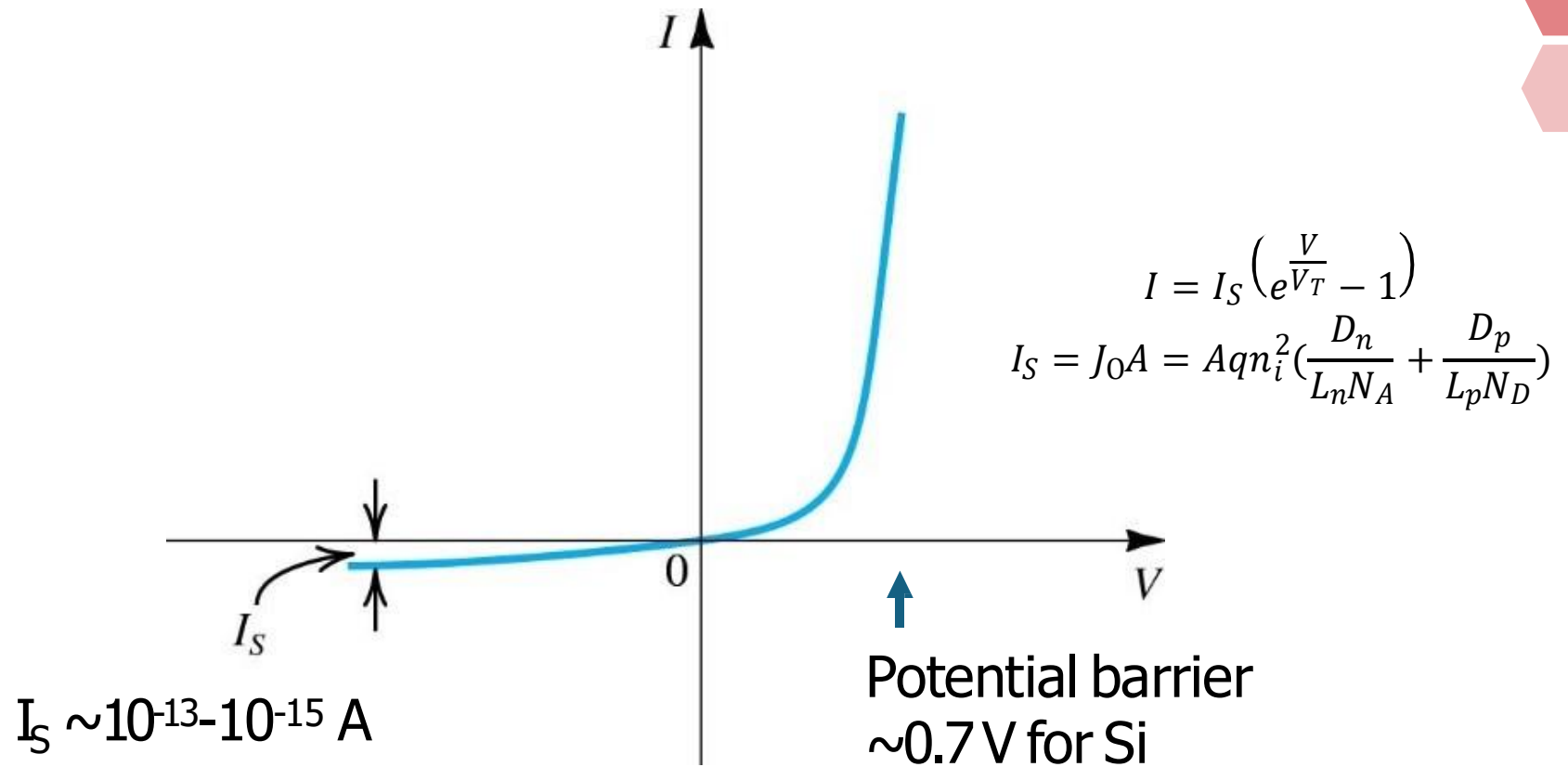
$$I = Aq\left(\frac{D_n}{L_n} n_{p0} + \frac{D_p}{L_p} p_{n0}\right)(e^{\frac{V}{V_T}} - 1)$$

given  $n_{p0} = \frac{n_i^2}{N_A}$   $p_{n0} = \frac{n_i^2}{N_D}$ , then:

$$I = \underbrace{Aqn_i^2\left(\frac{D_n}{L_n N_A} + \frac{D_p}{L_p N_D}\right)}_{I_S}(e^{\frac{V}{V_T}} - 1)$$

$I_S$  is your “saturation current”

# I-V relationship



More on the I-V curve next time...

# Example 1

What is the reverse saturation current at 398 K?

$$N_A = 10^{19} \text{ /cm}^3 \text{ in p-region}$$

$$N_D = 10^{17} \text{ /cm}^3 \text{ in n-region}$$

$$A = 1 \times 1 \text{ /um}^2$$

Assume that  $D_n$  and  $D_p$  follow these temperature distributions:

$$D_n = D_{n_0}(1 + 0.002(T - 300)), D_{n_0} = 36 \text{ cm}^2/\text{s}$$

$$D_p = D_{p_0}(1 + 0.002(T - 300)), D_{p_0} = 12 \text{ cm}^2/\text{s}$$

And  $L_n$  and  $L_p$  follow these temperature distributions:

$$L_n = L_{n_0}(1 - 0.001(T - 300)), L_{n_0} = 10 \text{ um}$$

$$L_p = L_{p_0}(1 - 0.001(T - 300)), L_{p_0} = 1 \text{ um}$$

$$n_i = BT^{\frac{3}{2}} e^{-\frac{E_g}{2kT}}$$

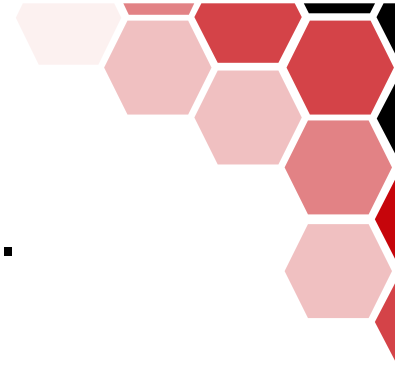
$$B = 7.3 \times 10^{15} \text{ cm}^{-3} \text{ K}^{-3/2}$$

$$k = 8.62 \times 10^{-5} \text{ eV/K}$$

$$I = I_S \left( e^{\frac{V}{V_T}} - 1 \right)$$

$$I_S = J_0 A = A q n_i^2 \left( \frac{D_n}{L_n N_A} + \frac{D_p}{L_p N_D} \right)$$

# Stored charge



Whenever we have an electric field, we have stored charge.  
How much is it???

On either side of the depletion region,  $Q$  is the same:

$$Q_J = qAx_nN_D = qAx_pN_A,$$

where  $A$  = cross-sectional area

We defined  $x_n$  and  $x_p$ , so

$$Q_J = qA \left( \frac{N_A N_D}{N_A + N_D} \right) w = A \sqrt{2\epsilon_s q \left( \frac{N_A N_D}{N_A + N_D} \right) V_0}$$

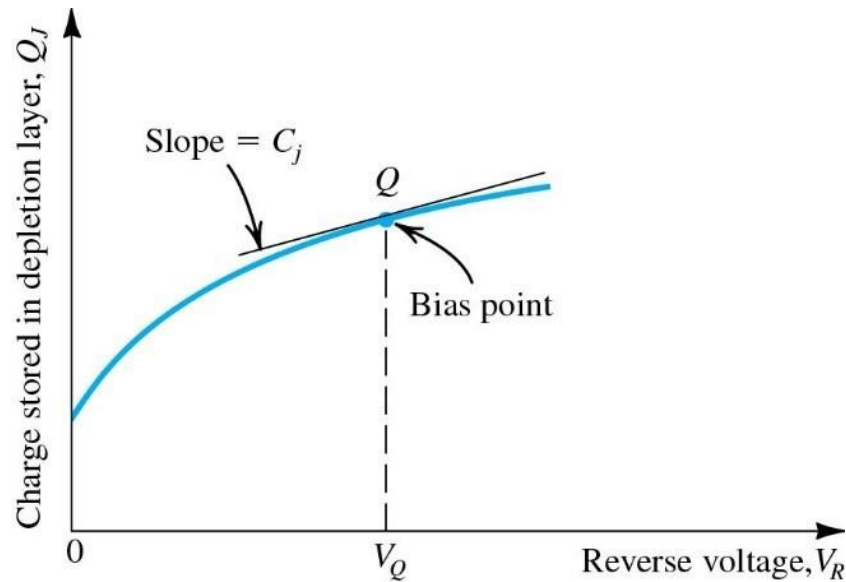




# Junction capacitance

In reverse bias, additional charges are stored in the depletion region:

$$Q_J = qA \left( \frac{N_A N_D}{N_A + N_D} \right) w = A \sqrt{2\epsilon_s q \left( \frac{N_A N_D}{N_A + N_D} \right) (V_0 + V_R)}$$



Because capacitance changes with  $V_R$ , it's hard to pin down a value

$$C_j = \frac{dQ_J}{dV_R} @V_R=V_Q$$

# Junction capacitance –graded junction

In the case where we do not have an abrupt junction, capacitance can be found as:

$$C_j = \frac{C_{j0}}{\left(1 + \frac{V_R}{V_0}\right)^m}$$

where

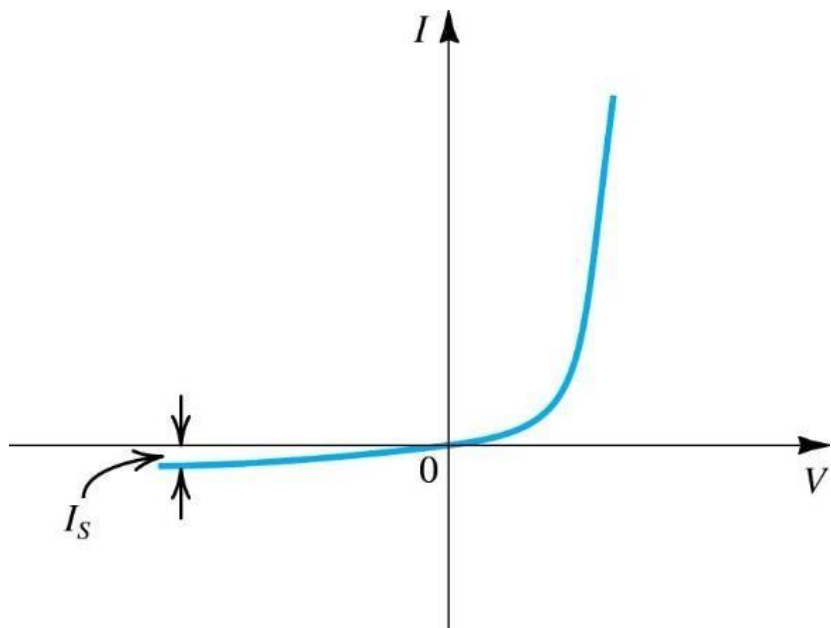
$$C_{j0} = A \sqrt{\frac{\epsilon_s q}{2} \left( \frac{N_A N_D}{N_A + N_D} \right) \left( \frac{1}{V_0} \right)}$$

and  $m$  is the grading coefficient, between 1/3 to 1/2

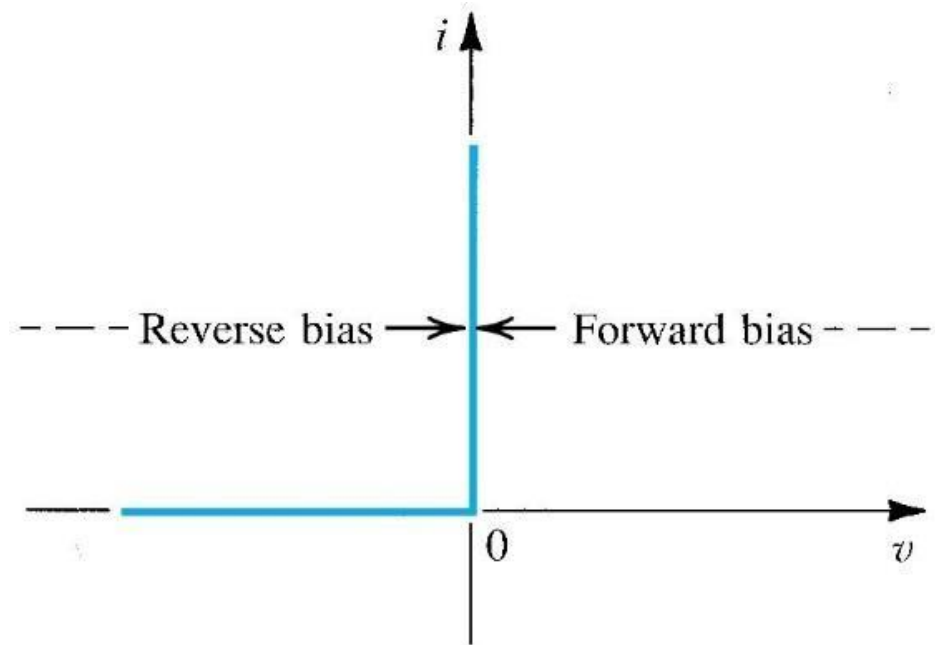
Capacitance will come up again when we enter the frequency domain. For now, we will move on.

# Chapter 4: Diodes





If I zoom out, the pn junction IV curve looks like this:



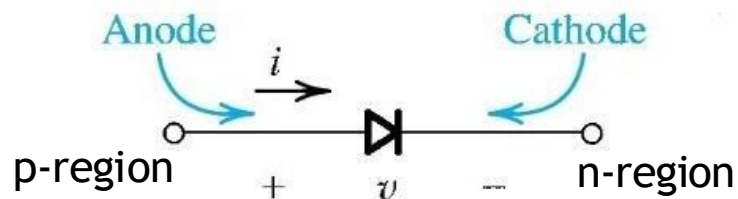
# The ideal diode model

Diode: Greek, two-way (two sides, fwd/up and reverse/down)

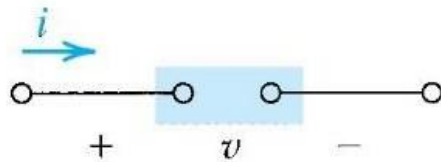
Anode: Greek, from ana 'up' + hodos 'way'

Cathode: Greek, from kata 'down' + hodos 'way'

## Circuit symbol:

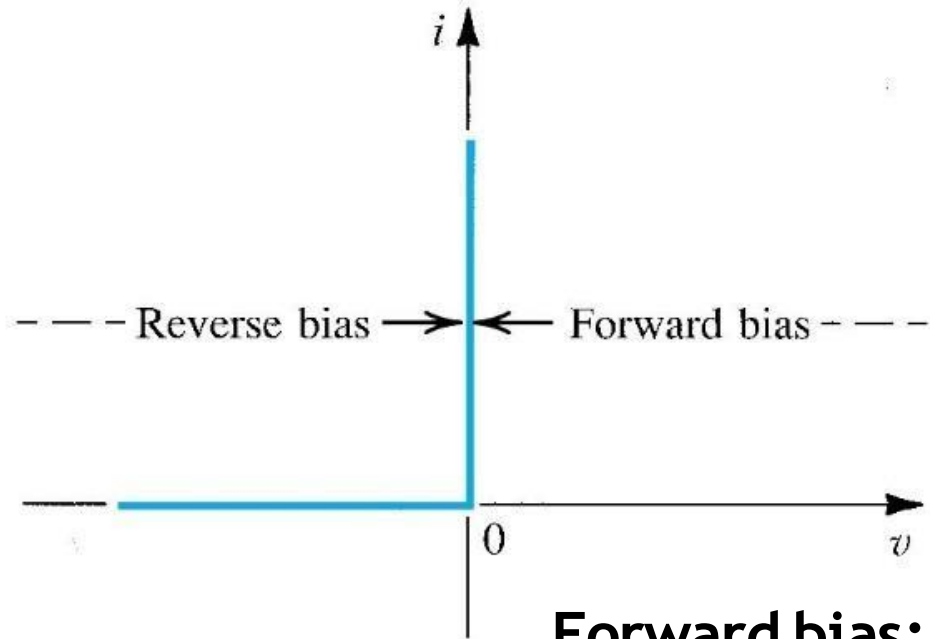
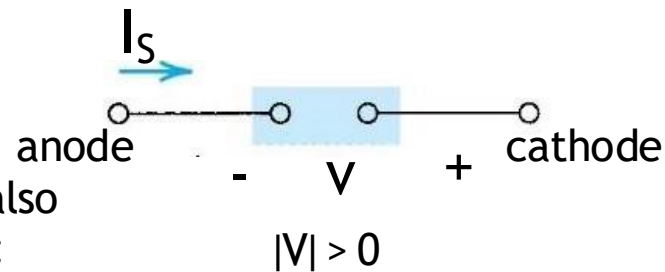


## Reverse bias:

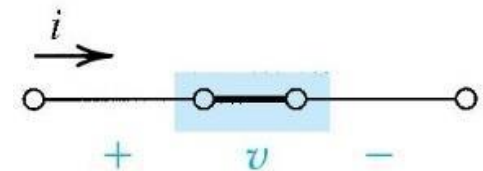


$$v < 0 \Rightarrow i = 0$$

Reverse bias also looks like this:



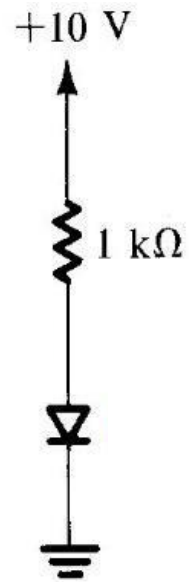
## Forward bias:



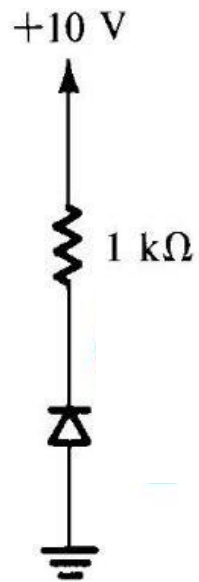
$$i > 0 \Rightarrow v = 0$$

# Examples

What is the current and voltage through/across the diodes?



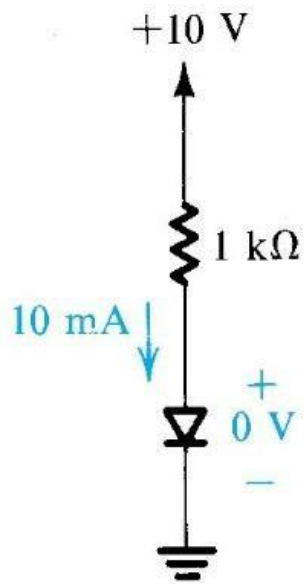
(a)



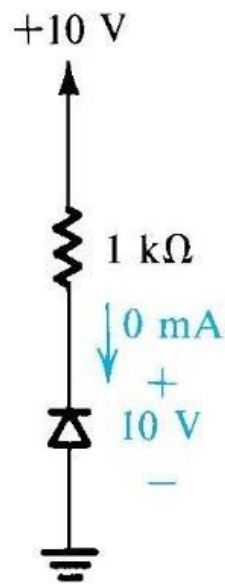
(b)

# Examples

What is the current and voltage through/across the diodes?



(a)

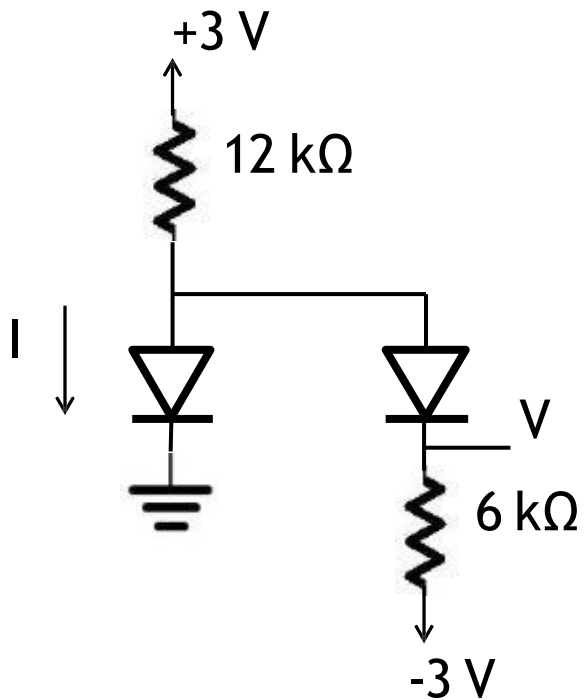


(b)

## Example 2

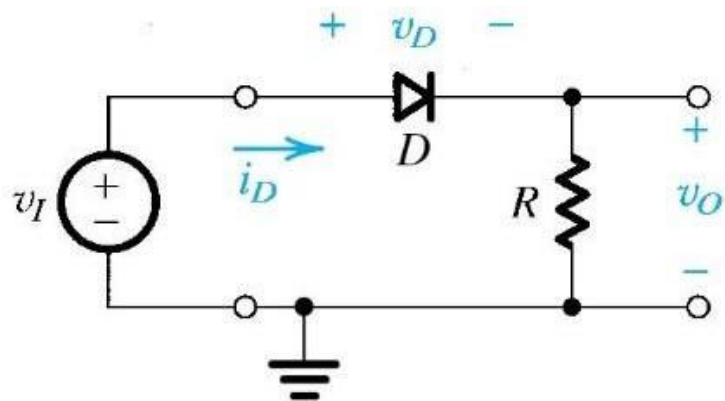
What about this circuit?

*Before proceeding with linear analysis, we have to know what operating state each diode is in!*

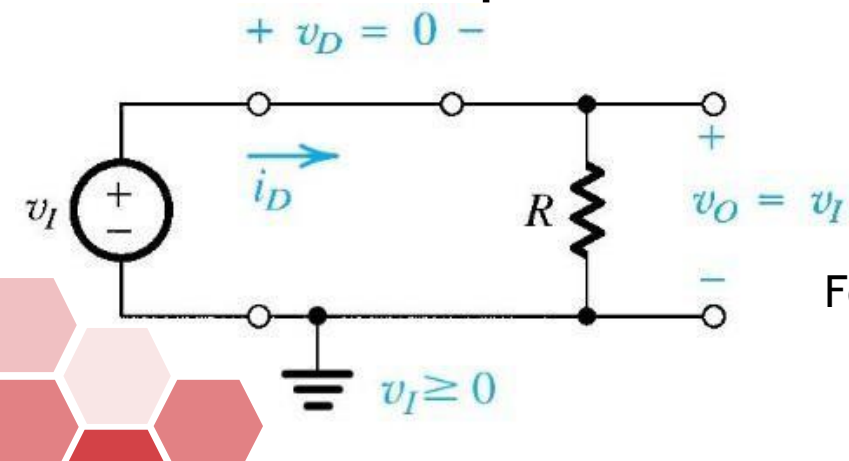




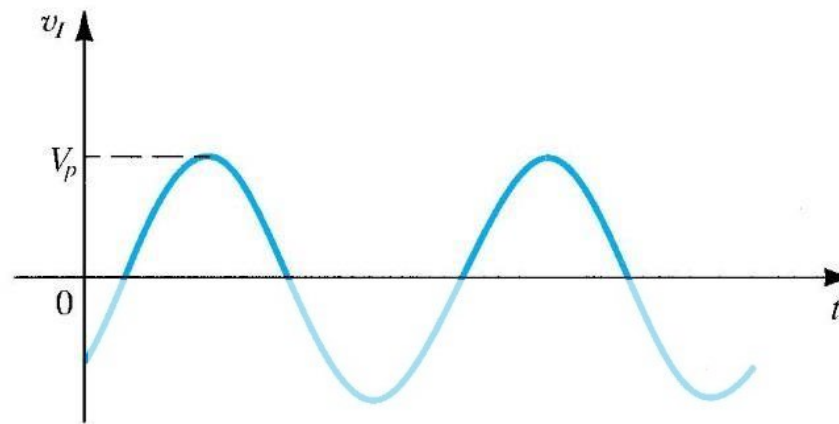
# Application: rectifier



Forward bias for positive  $v_I$



Let's say we have a sinusoid input, like from the wall:

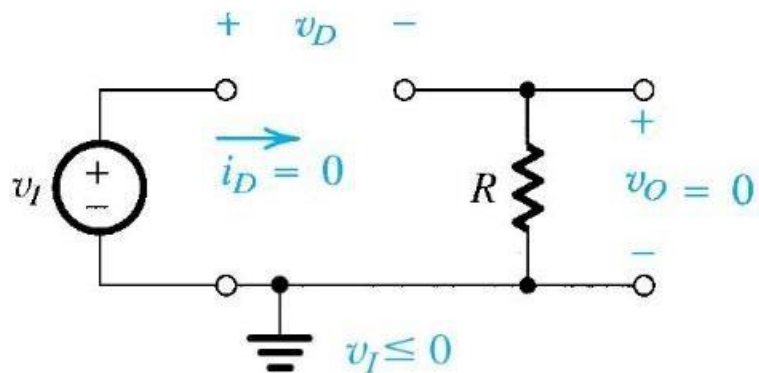


(b)

Focusing on just the positive part of the waveform

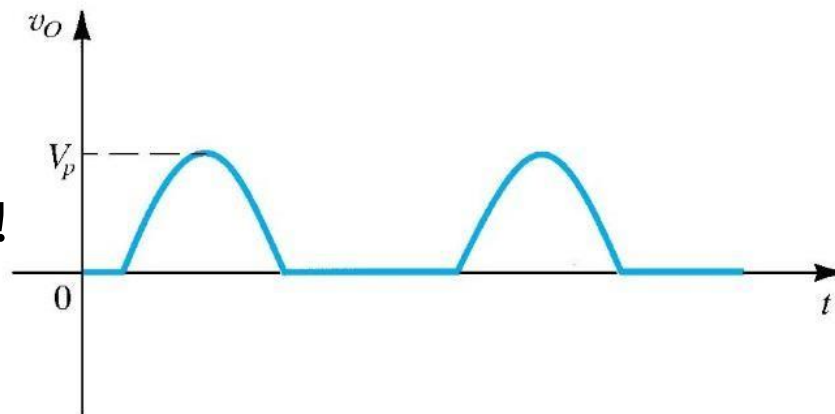
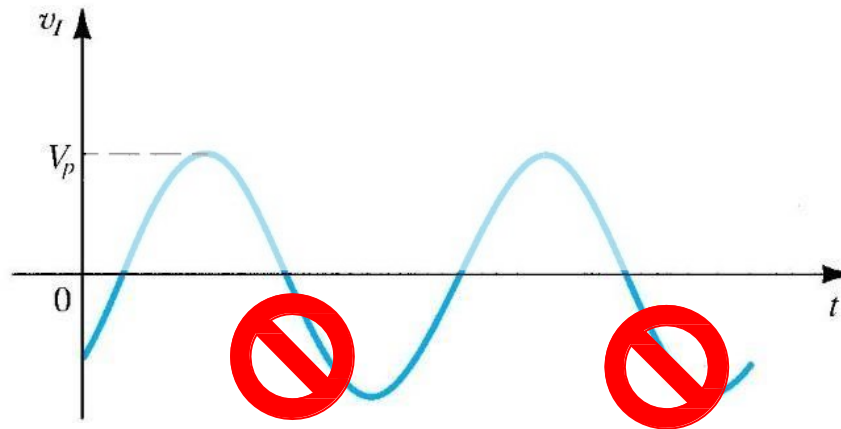
# Application: rectifier

Reverse bias for negative  $v_I$



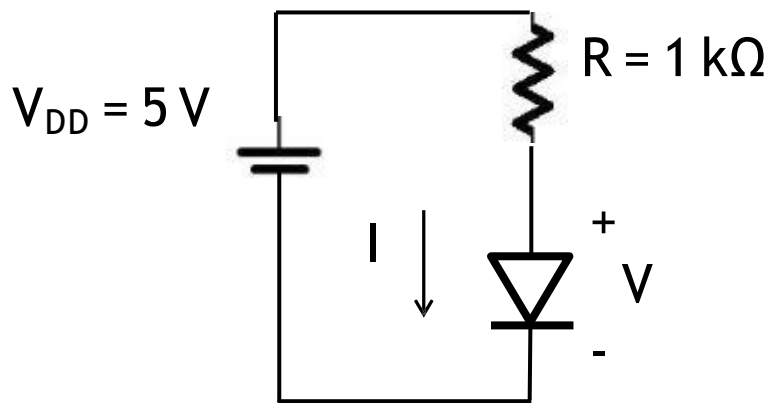
(d)

Output voltage has only one polarity!



(e)

## Example 3 – ideal model



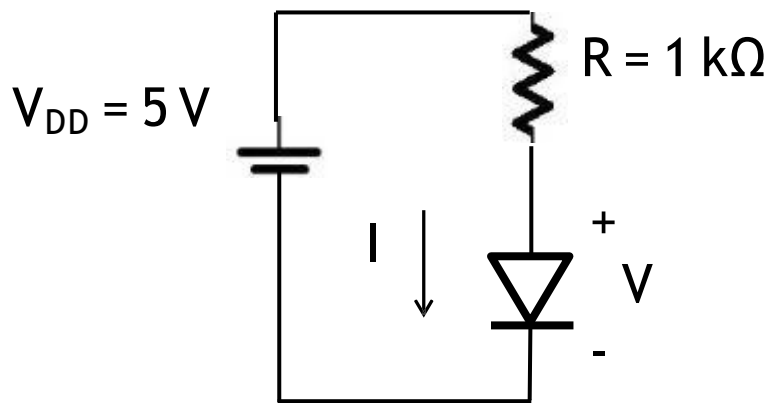
Ideal diode:

$$V = ?$$

$$I = ?$$

Diode is forward biased!  
Anode is at higher potential.

## Example 3 – ideal model



Ideal diode:

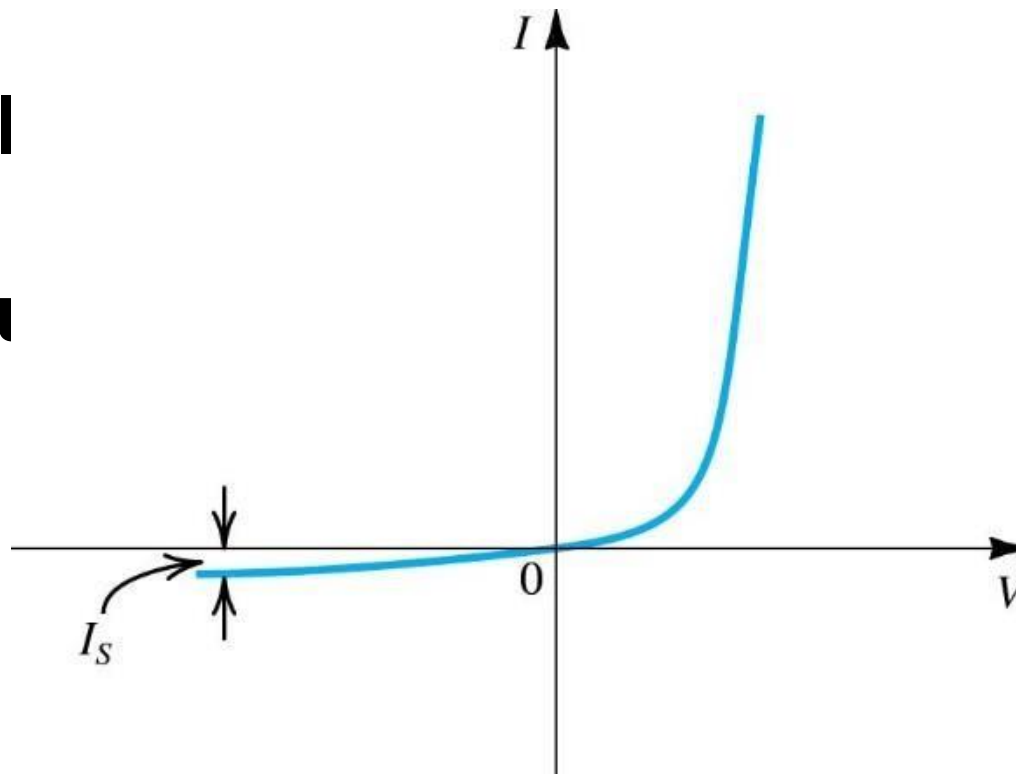
$$V = 0\text{ V}$$

$$I = 5\text{ V} / 1\text{ k}\Omega = 5\text{ mA}$$

Diode is forward biased!  
Anode is at higher potential.

**The ideal  
model is  
always si**

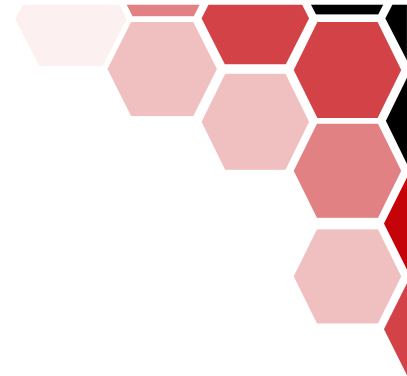
If I zoom in, the pn junction IV  
curve is exponential!



# Exponential model (iterative approach, non-approximate)

$$I = I_S \left( e^{\frac{V}{V_T}} - 1 \right) \quad \text{In forward bias}$$

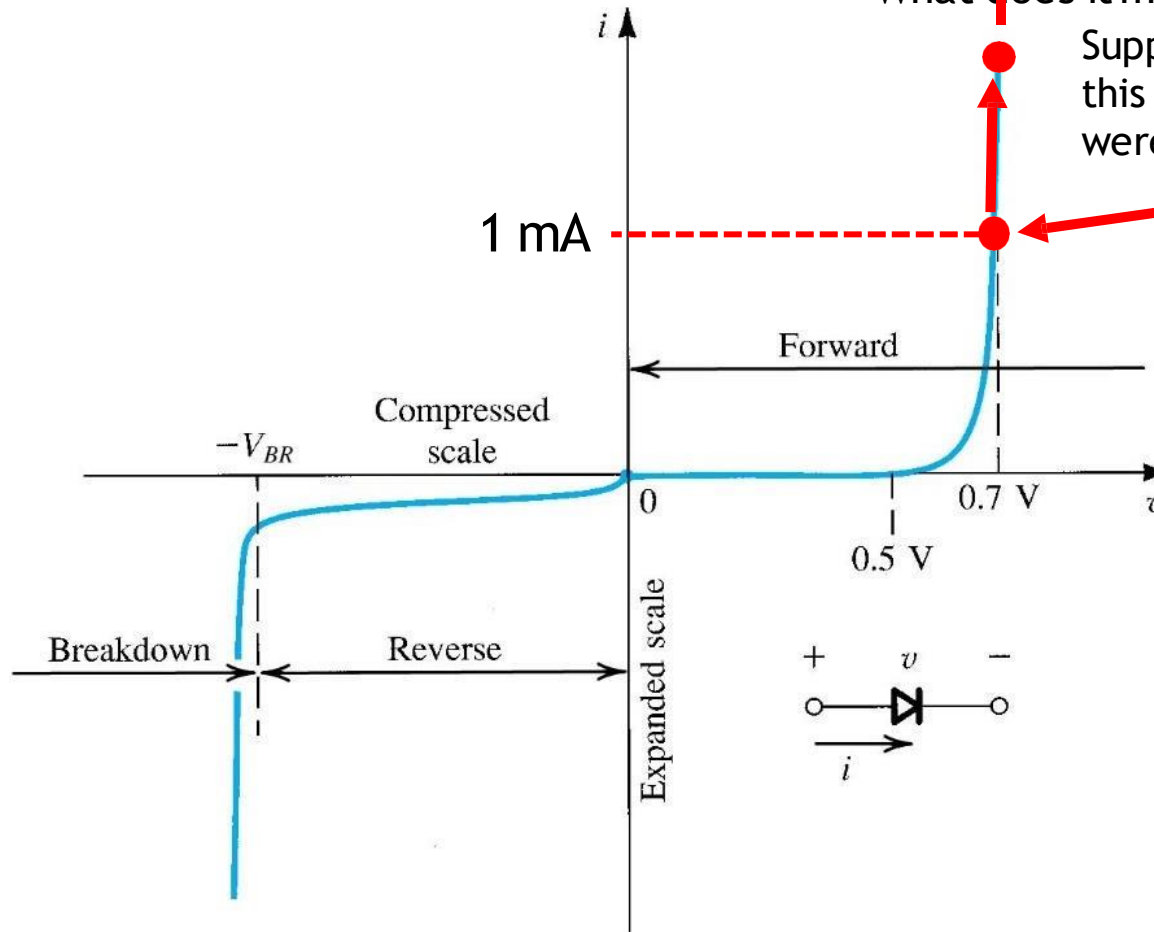
We can simplify:  $I \cong I_S \left( e^{\frac{V}{V_T}} \right)$  for  $I \gg I_S$



# Methodology to the iterative approach (non-approximate)

1. Start out with a known datapoint ( $I_D$  ,  $V_D$ ). (Use these values as a starting guess.)
2. Plug into the exponential equation.
3. Get new, more accurate values ( $I_D'$  ,  $V_D'$ ).
4. Plug it back into the exponential equation ( $I_D$  ,  $V_D$ ).
5. If these values ( $I_D''$  ,  $V_D''$ ) match the last values you calculated ( $I_D'$  ,  $V_D'$ ) (within some reasonable amount), you're done! (Otherwise, keep plugging C chugging.)

# I'm lost. What's going on?



Why do the exponential model?  
What does it mean?

Suppose you measured  
this diode and you  
were here

But say you put that  
same diode in a diff.  
circuit and want to know  
“where you are” AKA  
what your current and  
voltage values are.

The exponential model  
allows you to “trace the  
curve” using Bob’s  
reference values + your  
NEW circuit values

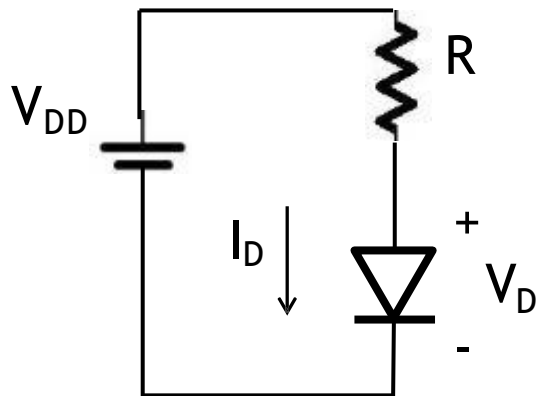
In other words, the iterative model  
lets you “step” along THIS curve until  
you CONVERGE on a solution



# Exponential model (non-approximate + book style)

Say that a fwd-biased diode has:  $I_D \cong I_S \left( e^{\frac{V_D}{V_T}} \right)$  corresponding to a diode voltage  $V_D$

using a  $V_D$  starting guess

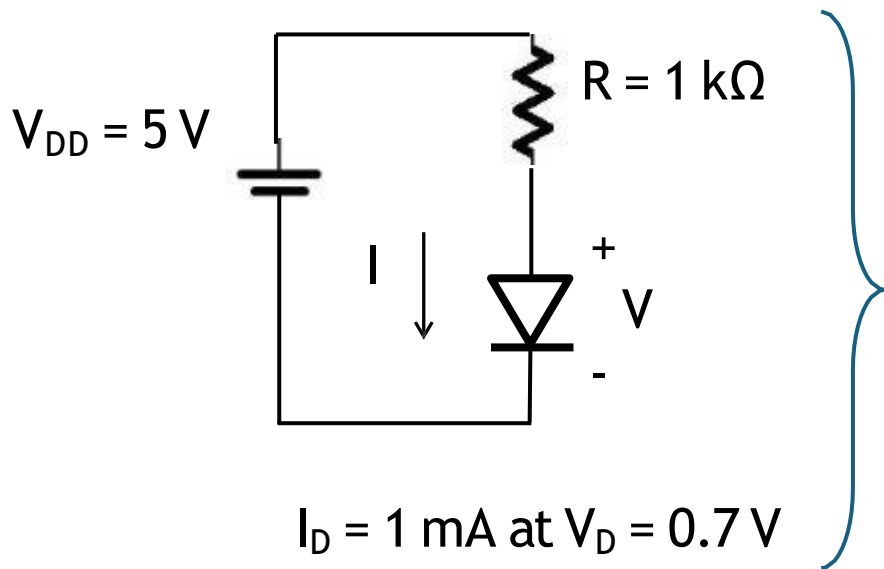


In a loop, we can get  $I_D$  from KCL:  $I_D = \frac{V_{DD} - V_D}{R}$  (Eq. 1)

Then we solve for  $V_D$ :  $V_D = V_T \ln \left( \frac{I_D}{I_S} \right)$  (Eq. 2)

And iterate over eq. 1 C eq. 2, alternating variables

# Example 3 – exponential model (non-approximate)



$$I_D = \frac{5\text{ V} - 0.7\text{ V}}{1\text{ k}\Omega} = 4.3\text{ mA}$$

Exponential model:

What is  $V$ ,  $I$ ?

$$\text{Start with } I_D = \frac{V_{DD} - V_D}{R}$$

$$V_D' = V_T \ln\left(\frac{I_D}{I_S}\right)$$

$$I = I_S \left( e^{\frac{V}{V_T}} - 1 \right)$$

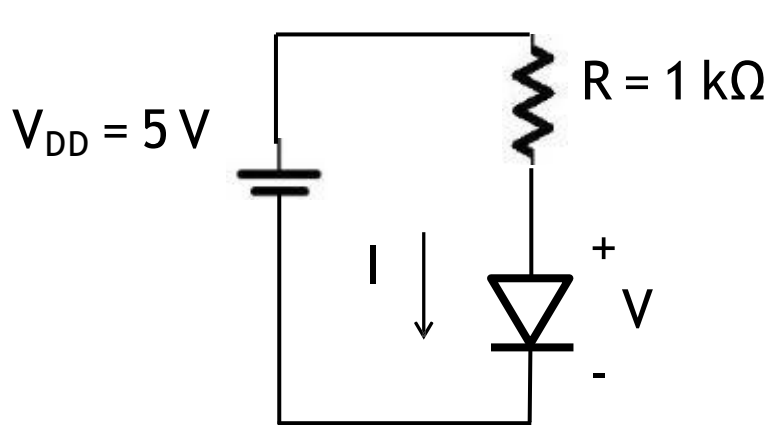
$$1\text{ mA} = I_S \left( e^{\frac{0.7\text{ V}}{25.9\text{ mV}}} - 1 \right) \Rightarrow I_S = 1.83 \times 10^{-15}\text{ A}$$

$$V_D' = 25.9\text{ mV} \ln\left(\frac{4.3\text{ mA}}{I_S}\right)$$

$$V_D' = 0.7378\text{ V}$$

Pay attention to # significant digits for accuracy!!

## Example 3 – exponential model (non-approximate)



$$I_D = 1 \text{ mA at } V = 0.7 \text{ V}$$

$$I_D'' = \frac{5 - 0.7375 \text{ V}}{1 \text{ k}\Omega} = 4.263 \text{ mA}$$

Close enough for 2 sig. digits!

Exponential model:

What is V, I?

$$\text{Start with } I_D = \frac{V_{DD} - V_D}{R}$$

$$V_D' = V_T \ln \left( \frac{I_D}{I_S} \right)$$

$$V_D' = 0.7378 \text{ V}, I_D = 4.3 \text{ mA}$$

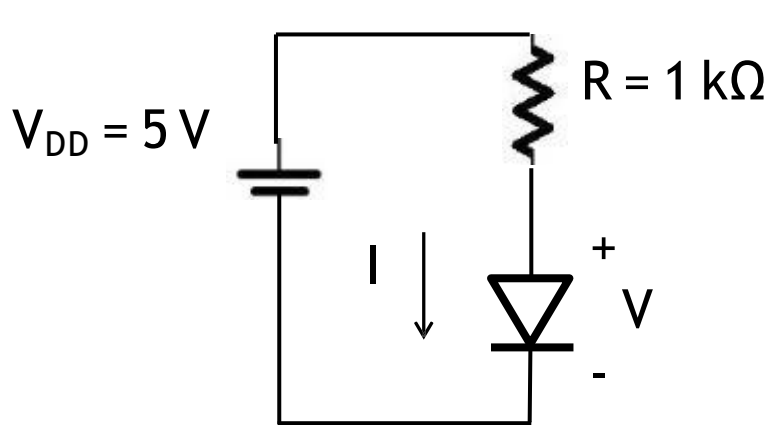
Not sure if this is really our final number! Iterate again to check

$$I_D' = \frac{5 - 0.7378 \text{ V}}{1 \text{ k}\Omega} = 4.2622 \text{ mA}$$

$$V_D'' = 25.9 \text{ mV} \ln \left( \frac{4.2622 \text{ mA}}{I_S} \right)$$

$$V_D'' = 0.7375 \text{ V}$$

## Example 3 – exponential model (non-approximate)



$I_D = 1 \text{ mA}$  at  $V = 0.7 \text{ V}$

$$I_D'' = \frac{5 - 0.7375 \text{ V}}{1 \text{ k}\Omega} = 4.26 \text{ mA}$$

$$V_D' = 0.74 \text{ V}$$

Exponential model:

What is  $V$ ,  $I$ ?

$$\text{Start with } I_D = \frac{V_{DD} - V_D}{R}$$

$$V_D' = V_T \ln \left( \frac{I_D}{I_S} \right)$$

$$V_D' = 0.7378 \text{ V}, I_D = 4.3 \text{ mA}$$

Not sure if this is really our final number! Iterate again to check

$$I_D' = \frac{5 - 0.7378 \text{ V}}{1 \text{ k}\Omega} = 4.2622 \text{ mA}$$

$$V_D'' = 25.9 \text{ mV} \ln \left( \frac{4.2622 \text{ mA}}{I_S} \right)$$

$$V_D'' = 0.7375 \text{ V}$$